European Commission

PRODUCT USER MANUAL

OMI_CLIMATE-VARIABILITY
Variability on average in the NINO 3.4 area
GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_SST_anom
GLOBAL_OMI_ CLIMATE-VARIABILITY_NINO34_Tzt_anom

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2.0	2022/03/30	All	Update of template	F Gues		Marie Drévillon
3.0	2024/06/30	All	Update with new product and extension year 2023	O. Le Galloudec		O. Le Galloudec

GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
ARC	Artic Ocean
CGLOR	C-GLORS reanalysis
CMEMS	Copernicus Marine Environment Monitoring Service
FWC	Freshwater Content
GLO	Global
GLOR	GLORYS2V4 reanalysis
GREP	Global Reanalysis Ensemble Product
NINO34	Nino 3.4 area in the Tropical Pacific 170°W-120°W 5°S-5°N
SST	Sea Surface Temperature

Table of Content

GLOSSARY AND ABBREVIATIONS **3**

DATA ACCESS **5**

I INTRODUCTION **5**

I.1. Summary **5**

I.2. History of changes **6**

II DESCRIPTION OF THE PRODUCT **7**

II.1. General information **7**

II.2. Details of datasets **8**

II.3. Product System Description **9**

II.3.1. GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_SST_anom 9

II.3.2. GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_Tzt_anom 10

III FILE FORMAT **10**

IV FILES NOMENCLATURE **10**

III REFERENCES **11**

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) family OMI_CLIMATE-VARIABILITY made from global product : it provides time series of climate variability indices such as the Sea Surface Tempature anomaly (sst_anom) and sub surface temperature anomaly (zt) averaged in the NINO3.4 area.

It contains 1D (time dimension) and 2D (depth time) anomaly information computed from GREP ensemble product of renanalysis. The time series are available over the 1993-2023 period. The temporal sampling is one month. The time series will be extended monthly (and closer to present time) starting in 2020.

GLOBAL_OMI_CLIMATE-VARIABILITY includes two products:

- global_omi_climate-variability_nino34_sst_anom contains monthly mean time series of SST anomaly computed over NINO3.4 Area for 1993 – 2023 period. This dataset is produced using GREP ensemble mean of three global reanalyses (GLOBAL_MULTIYEAR_PHY_ENS_001_031).
- global_omi_climate-variability_nino34_Tzt_anom contains monthly mean time series of sub-surface temperature anomaly computed over NINO3.4 Area for 1993 – 2023 period. This dataset is produced using GREP ensemble mean of three global reanalyses (GLOBAL_MULTIYEAR_PHY_ENS_001_031).

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The two OMIs are described in the Quality Information Document (QUID) CMEMS-OMI-QUID-CLIMATE-VARIABILITY-NINO34_Tzt_anom.pdf (copernicus.eu) and CMEMS-OMI-QUID-CLIMATE-VARIABILITY-NINO34_SST_anom.pdf (copernicus.eu) .

I.2. History of changes

2018/09	Publication of GLOBAL_OMI_CLIMATE-VARIABILITY _NINO34 _SST_anom and product GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34 _Tzt_anom (GLOBAL_OMI_CLIMATE-VARIABILITY family)
2022/03	Update
2024/11	Update with new product and extension year 2023

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_SST_anom, GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_Tzt_anom
Geographical coverage	NINO34 area
Variables	Sea Surface Temperature, subsurface temperature and density (sigma-theta)
Update frequency	Annual
Available time series	From 1993 to 2023
Delivery mechanism	Marine Data Store
Temporal/Spatial resolution	Monthly averages
Integration depth	Surface and subsurface
Format	NetCDF CF1.7

II.2. Details of datasets

GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_SST_anom contains the 1D monthly NINO34 SST anomalies : from GREP ensemble mean
sst_mean [degree C] Ensemble Mean of Sea Surface Temperature anomaly averaged in the NINO3.4 area from GREP reanalysis Sea_surface_temperature **sst_std [degree C]** Standard Deviation of Sea Surface Temperature anomaly averaged in the NINO3.4 area from GREP reanalysis Sea_surface_temperature **sst_glor [degree C]** Sea Surface Temperature anomaly averaged in the NINO3.4 area from GLORYS2V4 reanalysis Sea_surface_temperature **sst_cglo [degree C]** Sea Surface Temperature anomaly averaged in the NINO3.4 area from C-GLORS reanalysis Sea_surface_temperature **sst_oras [degree C]** Sea Surface Temperature anomaly averaged in the NINO3.4 area from ORAS5 reanalysis Sea_surface_temperature
GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_Tzt_anom contains the 2D monthly NINO34 sub-surface temperature anomalies and sigma-theta: from GREP ensemble mean
thetao_mean [degree C] Ensemble Mean of Sea Water Temperature anomaly averaged in the NINO3.4 area from GREP reanalysis sea_water_temperature **thetao_std [degree C]** Standard Deviation of Sea Water Temperature averaged in the NINO3.4 area from GREP reanalysis sea_water_temperature **thetao _glor [degree C]**

Sea Water Temperature anomaly averaged in the NINO3.4 area from GLORYS2V4 reanalysis Sea_water_temperature **thetao _cglo [degree C]** Sea Water Temperature anomaly averaged in the NINO3.4 area from C-GLORS reanalysis Sea_ water _temperature **thetao _oras [degree C]** Sea Water Temperature anomaly averaged in the NINO3.4 area from ORAS5 reanalysis Sea_ water _temperature **sigma_theta_mean [°C]** Ensemble Mean of Sea Water Temperature Sigma-Theta averaged in the NINO3.4 area from GREP reanalysis sea_water_sigma_theta **sigma_theta_std [degree C]** Standard Deviation of Sea Water Temperature Sigma-Theta averaged in the NINO3.4 area from GREP reanalysis sea_water_sigma_theta **sigma_theta_glor [degree C]** Sea Water Temperature Sigma-Theta averaged in the NINO3.4 area from GLORYS2V4 reanalysis sea_water_sigma_theta **sigma_theta_cglo [degree C]** Sea Water Temperature Sigma-Theta averaged in the NINO3.4 area from C-GLORS reanalysis sea_water_sigma_theta **sigma_theta_oras [degree C]** Sea Water Temperature Sigma-Theta averaged in the NINO3.4 area from ORAS5 reanalysis sea_water_sigma_theta

II.3. Product System Description

II.3.1. GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_SST_anom

This product is an indicator of the state of the central tropical Pacific el Niño conditions. It provides the sea surface temperature anomaly over the standard NINO 3.4 area (170°W - 120°W, 5°S – 5°). It is computed from GREP ensemble mean of global reanalyses. The anomaly is calculated compared to a 1993-2014 reference period also based on GREP (GLOBAL_MULTIYEAR_PHY_ENS_001_031). This indicator based on a multi-product approach is computed directly from the ensemble mean (it is not the ensemble mean of indicators computed from each of the members), the individual members are also provided in the file.
The monthly mean time series start on January 1993 and ends on December 2023.

II.3.2. GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_Tzt_anom

This product is an indicator of the state of the central tropical Pacific el Niño conditions. It provides the sub-surface temperature anomalies on average over the standard NINO 3.4 area (170°W - 120°W, 5°S – 5°).
It is computed from GREP ensemble mean of global reanalyses. The anomaly is calculated compared to a 1993-2014 reference period also based on GREP (GLOBAL_MULTIYEAR_PHY_ENS_001_031). This indicator based on a multi-product approach is computed directly from the ensemble mean (it is not the ensemble mean of indicators computed from each of the members). Sigma-theta computed from GREP over the same area is also provided. The same quantities computed from the individual members of GREP are also provided in the file.
The monthly mean time series start on January 1993 and ends on December 2023.

III FILE FORMAT

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

OMI_CLIMATE-VARIABILITY files nomenclature when downloaded through the CMEMS Marine Data Store Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>.nc

Where :

- **<datasetname>** is the dataset reference **(global_omi_health_carbon_co2_flux_integrated** or **global_omi_health_carbon_ph_area_averaged).**
- **<timestep>** is the first time step contained in the file**.**
- **<productiontime>** is the production time when the file was generated.
- **.nc** is the standard NetCDF filename extension.

Example: global_omi_climate-variability_nino34_sst_area_averaged_anomalies_1993_P20220427_R19932014.nc

III REFERENCES

See the Quality Information Document (QUID) and Product User Manual (PUM):
CMEMS-OMI-QUID-CLIMATE-VARIABILITY-NINO34_SST_anom.pdf (copernicus.eu)
CMEMS-OMI-QUID-CLIMATE-VARIABILITY-NINO34_Tzt_anom.pdf (copernicus.eu)
CMEMS-OMI-PUM-NINO34.pdf (copernicus.eu)